

RONAK KAMBOJ

Bhopal, Madhya Pradesh | 9244331984 | ronakkamboj26@gmail.com
LinkedIn: [linkedin.com/in/ronak-kamboj](https://www.linkedin.com/in/ronak-kamboj) | GitHub: github.com/r7rainz

Summary

Computer Science undergraduate at VIT Bhopal with hands-on experience building backend systems and full-stack web applications using Go, Node.js, React.js, Next.js, PostgreSQL, REST APIs, WebSockets, and Docker. Strong interest in system design, distributed systems, software architecture, and DevOps; seeking an IT or backend/web development internship to contribute to production-minded software and grow toward full-stack development.

Education

Bachelor of Technology in Computer Science | VIT Bhopal | 2023–2027 | CGPA: 8.56/10
Sagar Public School, Saket Nagar | Class XII (HSC): 85.2% (2023) | Class X (SSC): 92.4% (2021)

Technical Skills

- **Languages:** Go (Golang), C++, TypeScript, JavaScript, SQL
- **Frontend:** React.js, Next.js
- **Backend & APIs:** Node.js, Fastify, Express.js, REST APIs, WebSockets, Server-Sent Events (SSE), OAuth 2.0, Microservices
- **Databases & Messaging:** PostgreSQL, MySQL, Redis, Kafka, Prisma, Supabase, Firebase
- **DevOps & Infrastructure:** Docker, Docker Compose, Git, GitHub Actions, Linux
- **Systems & Core CS:** System Design, Distributed Systems, Data Structures & Algorithms, Operating Systems, DBMS

Projects

PulseOps | *Go, Node.js, Kafka, PostgreSQL, Redis, Next.js, Docker* | github.com/r7rainz/pulseops

- Built a self-hosted infrastructure monitoring platform using Fastify, Next.js, and PostgreSQL to monitor uptime, latency, and SSL certificate expiry.
- Developed a concurrent ping engine in Go using goroutine worker pools and Apache Kafka to process monitoring jobs asynchronously.
- Built a webhook alerting service with Redis, BullMQ, and Docker Compose, providing automatic retries, exponential backoff, and incident tracking across microservices.

AuraMail | *Go, Next.js, PostgreSQL, OpenAI GPT-4* | github.com/r7rainz/auramail

- Built an AI-powered placement email analyzer that connects to Gmail using Google OAuth 2.0 and automatically categorizes placement-related emails.
- Used OpenAI GPT-4 Mini to extract structured information including company, role, deadline, salary, and eligibility from unstructured email content.
- Built the backend in Go with Server-Sent Events (SSE) for real-time processing updates and stored application data in PostgreSQL using Goose migrations.

SongSpot | *Go, React, TypeScript, Redis, WebSockets, Docker* | github.com/r7rainz/songspot

- Built a collaborative music platform where users join rooms with six-character invite codes and stay synchronized while listening through embedded YouTube players.
- Developed a concurrent backend in Go using WebSockets, Redis Pub/Sub, and per-room hubs to synchronize playback events across connected clients with low latency.
- Implemented clock-offset estimation and timestamp-based playhead synchronization for participants with different network latency and system clocks.

Switchyard | *Go, Next.js, React Flow, PostgreSQL, WebSockets, Docker* | github.com/r7rainz/switchyard

- Built an AI-powered workflow automation platform that converts natural-language engineering tasks into editable workflow graphs and executes them with real-time status and logs.
- Developed a Go backend with a modular workflow engine, REST APIs, and WebSockets supporting asynchronous execution, retries, cancellation, idempotency, and execution recovery.
- Implemented workspace RBAC, encrypted credential storage, OAuth integrations, signed webhooks, SSRF protection, and integrations with GitHub, Slack, Discord, SMTP, and multiple LLM providers.

Achievements & Activities

- **Open Source Recognition:** Arch Linux rice featured on It's FOSS as one of the best Hyprland desktop configurations.
- **Competitive Programming:** Codeforces Pupil (1245 rating), actively solving algorithmic programming problems.
- **Technical Efficiency:** Ranked in the top 0.01% globally with a typing speed exceeding 150 WPM.
- **Competitions:** Secured 3rd place in the Technical Quiz at Tech Fest, Delhi Public School Kolar.
- **Athletics & Interests:** State-level badminton player, college football team member, football, and gym.